

Background Summary

Quantum Computing Concepts for Identifying Simulated Quantum Noise

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1. Purpose

Here's the quantum computing background the project needs. I go through qubits, gates and measurement quickly, then spend most of the time on noise, since that's what the project is really about: telling noise types apart from measurement results.

2. Qubits and superposition

A classical bit is just 0 or 1. A qubit is more flexible. Its state is written $\alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers with $|\alpha|^2 + |\beta|^2 = 1$, and that mix is what people call superposition. You can't read α and β off directly, though. Measuring gives a single 0 or 1: 0 with probability $|\alpha|^2$, 1 with probability $|\beta|^2$.

For one qubit there's a nice geometric picture called the Bloch sphere. The state is a point on (or inside) a sphere, with $|0\rangle$ at the top and $|1\rangle$ at the bottom. I use it later because it makes the three noise types easy to tell apart.

3. Gates, circuits and entanglement

Gates change a qubit's state. X swaps $|0\rangle$ and $|1\rangle$, like a NOT. H (Hadamard) takes $|0\rangle$ to an even mix of $|0\rangle$ and $|1\rangle$. Z flips the sign on the $|1\rangle$ part (a phase flip), which an ordinary 0/1 measurement can't see. CNOT acts on two qubits and flips the second only when the first is 1.

Apply H to the first qubit and then CNOT to a pair starting in $|00\rangle$ and you get the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$. Each qubit on its own looks like a coin flip, but the two results always match, which is entanglement. A circuit is just a list of gates ending in a measurement. Mine will stay small, two or three qubits.

4. Measurement

Each run of a circuit (a "shot") gives one outcome, so to estimate probabilities you run it many times and count. The counts are only estimates and they wobble. With N shots the uncertainty on a probability p is about $\sqrt{p(1-p)/N}$, so fewer shots give blurrier frequencies. The number of shots is one of the things I plan to vary, for that reason.

The basis matters too. A standard (Z-basis) measurement only tells you about 0 versus 1. To get at phase, you apply H just before measuring, which amounts to measuring in the X basis.

5. Quantum noise

Real devices aren't perfect. Gates are a bit off, qubits leak information into their surroundings, and readout sometimes reports the wrong value. Preskill (2018) calls this the NISQ era, short for noisy intermediate-scale quantum. A simulator like Qiskit Aer lets me add noise on purpose and control how strong it is, which is what makes this project doable. I'm looking at three standard single-qubit models.

Depolarizing noise. With some probability the qubit's state is replaced by a completely random one. On the Bloch sphere the point shrinks toward the centre by the same amount in every direction, so it shows up whichever basis you measure in.

Amplitude damping. This models energy loss, like an excited atom relaxing. $|1\rangle$ decays to $|0\rangle$ with some probability, so the point gets pulled toward $|0\rangle$. A qubit prepared in $|1\rangle$ gives extra 0s, while one prepared in $|0\rangle$ isn't touched.

Phase damping (dephasing). The qubit keeps its energy but loses the relative phase between $|0\rangle$ and $|1\rangle$. A qubit sitting in $|0\rangle$ or $|1\rangle$ looks completely fine in the Z basis. The damage only shows in superposition states, and only if you measure in the X or Y basis. On the Bloch sphere the point squeezes in toward the vertical axis.

Figure 1 shows all three acting on the same starting state.

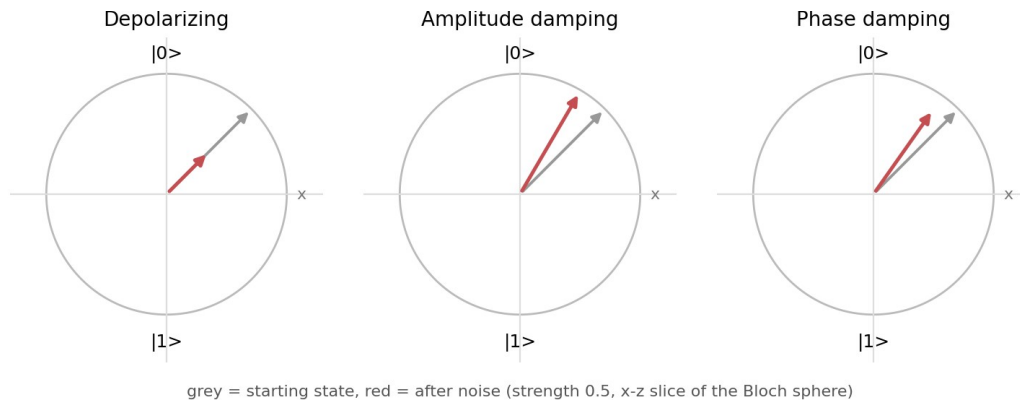


Figure 1. A starting state (grey) and the same state after each noise model (red), shown on the x-z slice of the Bloch sphere at noise strength 0.5.

6. A small experiment

To check I'd understood this, I ran two single-qubit circuits in Qiskit Aer under each noise model, with error probability 0.2 and 8,192 shots. In setting A, I prepare $|1\rangle$ and measure in the Z basis, so the ideal result is always 1. In setting B, I prepare $|+\rangle$ and measure in the X basis, so the ideal result is always 0. Figure 2 shows the fraction of wrong outcomes.

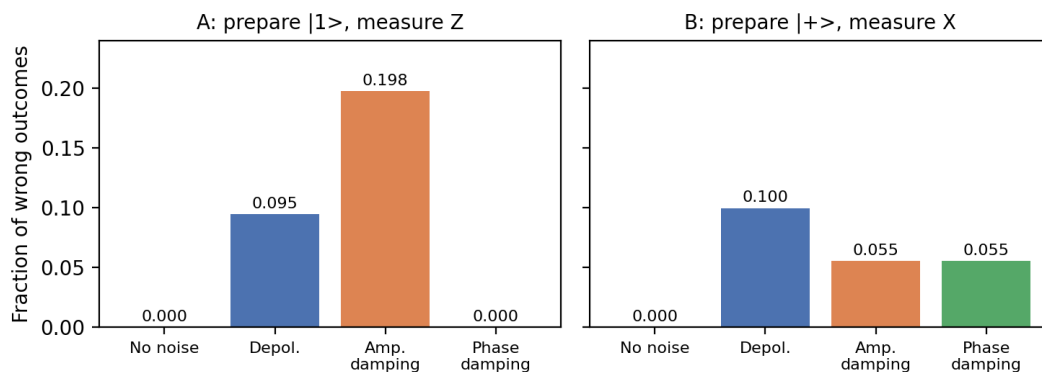


Figure 2. Fraction of wrong outcomes for each noise model in setting A (left) and setting B (right).

With no noise both settings give zero errors, as they should. In A, amplitude damping gave about 20% wrong outcomes, depolarizing about 10%, and phase damping none at all. The depolarizing and amplitude damping numbers match what the theory predicts ($0.2/2 = 0.1$, and 0.2). In B, depolarizing gave about 10% errors, and amplitude damping and phase damping both gave about 5.5%.

One thing I noticed in the results: neither setting separates all three noise types on its own. B can't tell amplitude damping from phase damping, and A can't tell phase damping from no noise at all. Used together they do separate all three, which is why the project uses several preparation and measurement settings.

7. Link to the project

The project is basically this experiment scaled up. I'll attach each noise model to the gates of small circuits, run many shots under several settings, and record the outcome frequencies. Those frequencies become the features the classifiers learn from, and the noise strength and the number of shots control how hard the problem is. The details go in the experimental plan.

8. Sources and learning materials

Watrous, J. Basics of Quantum Information. IBM Quantum Learning course (learning.quantum.ibm.com). Covers qubits, gates and measurement.

Qiskit Aer documentation, noise module (NoiseModel, depolarizing_error, amplitude_damping_error, phase_damping_error). Used for how the noise models are built and applied in simulation.

Nielsen, M. A., & Chuang, I. L. (2010). Quantum Computation and Quantum Information (10th anniversary ed.). Cambridge University Press. Chapter 8 on quantum noise and quantum operations.

Preskill, J. (2018). Quantum computing in the NISQ era and beyond. Quantum, 2, 79.

Code for Figures 1 and 2: make_bloch.py and quantum_noise_demo.py (included with this submission).